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No
Date

Tugas 2 | Deep Learning

1. Diketahui Input $x_1 = 0.01$ $t_1 = 0.79$ Learning rate = 0.5
 $x_2 = 0.52$ $t_2 = 0.21$

Fungsi aktivasi (sigmoid) : $\sigma(x) = \frac{1}{1 + e^{-x}}$

turunan sigmoid :

$$\frac{d}{dx}(\sigma(x)) = \sigma(x)(1 - \sigma(x))$$

$$\text{fungsi error : } E_{\text{total}} = \frac{1}{2} (t_1 - o_1)^2 + \frac{1}{2} (t_2 - o_2)^2$$

2. Forward pass (hidden layer)

Substitusi nilai :

$$\text{net } h_{11} = w_{11}x_1 + w_{12}x_2 + w_{b1}$$

$$= 0.15(0.01) + 0.20(0.52) + 0.35 = 0.4555$$

Output

$$o_{h11} = \sigma(0.4555) = \frac{1}{1 + e^{-0.4555}} = 0.6119$$

$$\text{net } h_{12} = 0.25(0.01) + 0.30(0.52) + 0.35 = 0.5085$$

$$o_{h12} = 0.62499$$

Output layer

$$\text{net } o_{21} = w_{21}o_{h11} + w_{22}o_{h12} + w_{b2}$$

$$= 0.40(0.6119) + 0.95(0.62499) + 0.60 = 1.12579$$

$$o_{o21} = 0.75505$$

$$\text{net } o_{22} = 0.50(0.6119) + 0.55(0.62499) + 0.60$$

$$= 1.29999$$

$$o_{o22} = 0.77721$$

3. Hitung total error

$$\sum \text{total} = \sum \frac{1}{2} (\text{target} - \text{output})^2$$

$$E_1 = \frac{1}{2} (0.79 - 0.75505)^2 = 0.0004$$

$$E_2 = \frac{1}{2} (0.21 - 0.77721)^2 = 0.16086$$

$$E_{\text{total}} = 0.16127 \approx 0.161$$

4. Backpropagation - output layer

$$\frac{\partial E_{\text{total}}}{\partial w_{21}} = \frac{\partial E_{\text{total}}}{\partial \text{out}_{021}} \cdot \frac{\partial \text{out}_{021}}{\partial \text{net}_{021}} \cdot \frac{\partial \text{net}_{021}}{\partial w_{21}}$$

• turunk error terhadap output

$$\frac{\partial E_{\text{total}}}{\partial \text{out}_{021}} = -(1 - \text{out}_{021}) = -0.75505 \approx -0.03995$$

• turunk sigmoid out

$$\frac{\partial \text{out}_{021}}{\partial \text{net}_{021}} = \text{out}_{021} (1 - \text{out}_{021}) = 0.75505 (1 - 0.75505) = 0.18086$$

• turunk net terhadap bobot

$$\frac{\partial \text{net}_{021}}{\partial w_{21}} = \text{out}_{011} = 0.61199$$

• gabungkan

$$\frac{\partial E_{\text{total}}}{\partial w_{21}} = 0.03995 \cdot 0.18086 \cdot 0.61199 = 0.00391$$

• update bobot

$$w_{21} = w_{21} - \eta \frac{\partial E}{\partial w_{21}}$$

$$= 0.90 - 0.5 (0.00391) = 0.90197$$

5. Backpropagation - Hidden layer

karena neuron hidden mempengaruhi dua neuron output

$$\Delta E_{total} = \Delta E_{o21} + \Delta E_{o22}$$

$$\Delta out_{h1} \quad \Delta out_{h1} \quad \Delta out_{h1}$$

$$= \left(\frac{\Delta E_{o21}}{\Delta net_{o21}} \cdot \frac{\Delta net_{o21}}{\Delta out_{h1}} \right) + \left(\frac{\Delta E_{o22}}{\Delta net_{o22}} \cdot \frac{\Delta net_{o22}}{\Delta out_{h1}} \right)$$

substitusi nilai

$$= (0.0395 \cdot 0.18986) (0.90) + (-0.56721 \cdot 0.1758)$$

$$(-.50)$$

$$= -0.09651$$

turunan terhadap w_{11}

$$\frac{\Delta E_{total}}{\Delta w_{11}} = \frac{\Delta E}{\Delta out_{h1}} \cdot \frac{\Delta out_{h1}}{\Delta net_{h1}} \cdot \frac{\Delta net_{h1}}{\Delta w_{11}}$$

$$\frac{out_{h1}}{net_{h1}} = 0.61199 (1 - 0.61199) = 0.2370$$

$$\frac{\Delta net_{h1}}{\Delta w_{11}} = x_1 = 0.01 = -0.00011$$

$$w_{11}^{baru} = 0.15 - 0.5 (-0.00011)$$

update bobot

$$v_{11} = 0.15 - 0.5 (-0.00011)$$

$$= 0.150055$$

hidden layer

$$w_{12} = 0.29994$$

$$w_{13} = 0.19715$$

$$w_{14} = 0.29586$$

$$w_{15} = 0.39991$$

output layer

$$w_{21} = 0.39396$$

$$w_{22} = 0.90197$$

$$w_{23} = 0.96997$$

$$w_{24} = 0.45201$$

$$w_{25} = 0.51927$$

$$w_{26} = 0.60322$$

$$w_{27} = 0.55091$$

turunan terhadap w_{22}

$$\frac{\partial E_{total}}{\partial w_{22}} = \frac{\partial E_{total}}{\partial out_{022}} \cdot \frac{\partial out_{022}}{\partial net_{022}} \cdot \frac{\partial net_{022}}{\partial w_{22}}$$

• turunan error terhadap output

$$\begin{aligned} \frac{\partial E_{total}}{\partial out_{022}} &= -(t_2 - out_{022}) \\ &= -(0.21 - 0.77721) = -(-0.56721) = 0.56721 \end{aligned}$$

• turunan sigmoid:

$$\begin{aligned} \frac{\partial out_{022}}{\partial net_{022}} &= out_{022} (1 - out_{022}) \\ &= 0.77721 (1 - 0.77721) = 0.17308 \end{aligned}$$

• turunan net terhadap input

$$\frac{\partial net_{022}}{\partial w_{22}} = out_{h11} = 0.61199$$

• gabungan

$$\frac{\partial E_{total}}{\partial w_{22}} = (-0.56721) \cdot 0.17308 \cdot 0.61199 = -0.06006$$

gabungan update

$$w_{22}^{baru} = 0.50 - 0.5 (-0.06006) = 0.9997$$

turunan terhadap w_{23}

$$\frac{\partial net_{021}}{\partial w_{23}} = out_{h12} = 0.62999$$

$$\frac{\partial E}{\partial w_{23}} = 0.03995 \cdot 0.18986 \cdot 0.62999 = 0.00902$$

$$w_{23}^{baru} = 0.95 - 0.5 (0.00902) = 0.95201$$

• nilai total error

$$E_{total} = 0.66197$$

Kesimpulan

Setelah dilakukan satu kali iterasi (epoch-1), nilai error sebesar 0.66197 dan seluruh bobot mengalami perubahan sesuai arah negatif gradien. Hal ini menunjukkan bahwa proses pembelajaran dengan backpropagation melalui bergerak menuju konvergensi untuk meminimalkan error.